

ParaEMT EMT-validation line (IAS2026 / TX4)

EMT Scientific Qualification Campaign (EMT-SQ1) — report

Branch `research/parent-emt-validation` (not pushed)

2026-09-12

Status: SQ1 STOPPED at SQ1-2 (blind synthetic qualification). By the preregistered rule, the ParaEMT EMT line is STOPPED PERMANENTLY.

- **SQ1-2** was run with the frozen V3 estimator (seed 20260913, 210 domain cases).
 - **Gate GB FAILS:** 78.4 % of the cases with $|\alpha| \geq 0.01$ were resolved, against ≥ 95 % required.
 - **Gate GC FAILS:** the p95 of $|\hat{\alpha} - \alpha|$ was 0.0066 s^{-1} , against ≤ 0.005 required.
 - **GA, GD and GE pass:** zero wrong-sign verdicts; p95 $|\hat{f} - f|$ of 0.0017 Hz ; no crash.
- **Not run:**
 - the phasor-TDS holdout T1–T9;
 - every device gate (the blind SG38, GFL35 and GFL37 were never simulated);
 - EMT04–EMT10.

1 History (binding; nothing rewritten)

campaign	preregistration	permanent outcome
V1	c2947bd8 (+A1)	G1 PASS; G3 FAIL, G4 FAIL (interface / device gates)
V2	0ab8efd4	SW PASS; G3a FAIL (caused by the frozen v1 estimator)
V3	6d339006 (+AM1)	E0 FAIL (criterion A5: single-channel ringdowns, 14/15)
SQ1	0555f587	SQ1-2 FAIL (GB 78.4 %, GC 0.0066) $\Rightarrow$ ParaEMT line stopped permanently

Statement. Four preregistered campaigns attempted an EMT corroboration of the IEEE-39 results.

- The first three stopped at infrastructure or measurement gates.
- The final, task-specific qualification campaign froze the estimator and the model interfaces in advance. It tested the estimator on a fresh blind synthetic set derived from the frozen phasor models of the actual scientific domain.
- The estimator made no wrong-sign call. It did not reach the preregistered resolution and accuracy targets. The failures concentrate where the target mode is a minority mode beside a dominant, similarly damped neighbouring mode family.
- By the preregistered rule, the ParaEMT line stopped before any blind device or portfolio holdout was simulated.

No EMT evidence for or against the portfolio, policy or design claims exists. None of the four failures is described as invalid.

2 SQ1-0 — frozen instrument

- **Estimator:** `experiments/parent_emt/v3/emtv3_estimator.py`, commit `f8e0d44f`, sha256 `849472d8...3b0817f`. It was unchanged throughout SQ1, and every SQ1 script asserts the hash.
- **Interfaces:** the v1 SG kernel and the V2 GFL voltage interface, frozen; no damping.

3 SQ1-1 — scientific measurement domain (frozen predictions only)

- **Target modes:** taken from the frozen `band_re`/`band_hz` of all 126 EMT04–EMT16 cases.
 - Hull: $\alpha \in [-0.229, +0.199] \text{ s}^{-1}$ and $f \in [0.557, 1.047] \text{ Hz}$.
 - Domain: $\alpha \in [-0.23, +0.20] \text{ s}^{-1}$ and $f \in [0.55, 1.05] \text{ Hz}$.

- **Linear modal library:** 26 frozen non-holdout phasor models (`results/EMTSQ1/domain/`).
 - It gives the channel residues of every transverse mode for the preregistered pulse, through the v1 §5 channel map.
 - Validation against the development TDS trace P4 BASE: 0.30 % median and 0.83 % maximum relative error.
 - 28–32 channels.
 - Per-channel target energy share: median 0.94, with 30 % of the channels below 0.1.
 - 16 nuisance modes of at least 1 % share.
- **Minority target.** In 9–10 of the 26 models, the preregistered target (the rightmost in-band mode, the 0.91-Hz family) is a *minority* mode.
 - A 0.64-Hz family lies 0.26 Hz away, decays only $0.003\text{--}0.07\text{ s}^{-1}$ faster, and carries 85–92 % of the median channel’s energy.
 - These models are the P4 subsets that contain 30, 35 or 37 without the full H4.
- **V3 S6** remains a stress test that documents a limitation of the estimator. It is not relabelled.

4 SQ1-2 — blind synthetic qualification (seed 20260913)

Generator. Each case takes a random library template.

- The template keeps its residues, perturbed by $\times U(0.8, 1.25)$ and $e^{jU(-0.2, 0.2)}$, and all of its nuisance modes.
- The target is redrawn from the domain.
- Affine trends and noise are added, with $\sigma = \nu \cdot \text{rms}$ and $\nu \sim \log U[10^{-4}, 3 \cdot 10^{-3}]$.
- $f_{\text{pred}} = f + U(\pm 0.03)$.
- Records are 90 s long; the frozen tracker, sentinel and adaptive rule are used.

stratum	n	resolved	correct concl.	wrong sign	p95 $ \Delta\alpha $	p95 $ \Delta f $
SD1 domain	130	82.3 %	80.8 %	0	0.0080	0.0022
SD2 near boundary ($ \alpha \leq 0.01$)	40	90.0 %	60.0 % [†]	0	0.0014	0.0003
SD3 low observability	40	67.5 %	67.5 %	0	0.0052	0.0014
ST1 stress (V3-S6-like; reported only)	20	100 %	100 %	0	0.0044	0.0010
ST2 stress ($\alpha \in [-0.6, -0.3]$; reported only)	15	0 %	0 %	0	—	—

[†] The rest are UNRESOLVED, which the prereg allows near the boundary.

gate (SD1–SD3)	rule	result	verdict
GA	0 wrong-sign verdicts for $ \alpha \geq 0.005$	0	pass
GB	$\geq 95\%$ resolved for $ \alpha \geq 0.01$	131/167 = 78.4 %	FAIL
GC	p95 $ \Delta\alpha \leq 0.005$	0.0066	FAIL
GD	p95 $ \Delta f \leq 0.005$	0.0017	pass
GE	no crash	none	pass

- **Calibration:** CI coverage of α_{true} among resolved cases is 94.7 %; 16 resolved cases have $|\Delta\alpha| > 0.005$.
- **Determinism rerun:** identical code and seed; the CSV is byte-identical (sha256 30f71465...4ab62f, 245 rows) and the JSON is identical except `wall_s` (`results/EMTSQ1/SQ1-2/sq1_rerun_identity.json`).

SQ1-2: FAIL $\Rightarrow$ STOP the ParaEMT line permanently (prereg SQ1 §3).

5 Post hoc diagnosis (not a gate)

Source: `sq1_posthoc_breakdown.json`. Domain cases with $|\alpha| \geq 0.01$:

template class	n	resolved	p95 $ \Delta\alpha $	p95 $ \Delta f $
dominant-nuisance (9 P4 subsets; the 0.64-Hz family carries $> 50\%$ of the median share)	54	40.7 %	0.059	0.127
all other templates	113	96.5 %	0.0056	0.0015

Why the 36 cases are unresolved:

- 13: estimate at the bound and CI too wide;
- 11: CI too wide only;

- 12: residual rule R3.

Reading.

- The frozen estimator is a single-mode targeted tracker, as the V3 preregistration specified.
- **Isolated or dominant target:** resolution $\approx 96.5\%$, with no wrong sign. Accuracy is marginal: $p95\ 0.0056\text{ s}^{-1}$, against the 0.005 target.
- **Minority target** beside a dominant, similarly damped mode 0.26 Hz away:
 - the estimator cannot isolate the target within the effective record length;
 - the jackknife guard withholds most of these cases (UNRESOLVED);
 - the remaining cases are biased.
- This is exactly the structure of about half the EMT05 P4 subsets. The failure is therefore *task-relevant, not an artefact*: an EMT05 run with this instrument could not have established $H_{\text{EMT}}(P4) = \{\{30, 33, 35, 37\}\}$.
- **Stress strata:** ST1 20/20. ST2 0/15, with no wrong sign: the CI is too wide for strongly damped multi-channel modes.

6 What was not run (and now never will be, in this line)

item	status
SQ1-3 phasor-TDS holdout T1–T9	NOT RUN; traces never generated
G3a/G3b (SG30, SG36, SG38 blind)	NOT RUN; SG38 never simulated
G4a/G4b (GFL30, GFL35, GFL37 blind)	NOT RUN; the V2 voltage interface was never run in a network
EMT04, EMT05, EMT06, EMT07/08, EMT10; the decision gate; EMT09, EMT11–EMT16	NOT RUN

Prepared but never executed: `sq1/EMTSQ1_refs.py`, `sq1/EMTSQ1_tds.py`, `sq1/EMTSQ1_gates.py` and `sq1/EMTSQ1_science.py`. They are committed with “never executed” banners and guarded by assertions on the preceding gates.

7 Final state of the EMT evidence

- **Network and operating point (V19/I01):** EMT QUANTITATIVELY REPRODUCED (V1 G1). This is the only EMT result.
- **SG and GFL device equivalence in EMT:** not established by any gate. The post hoc diagnostics (V1 D1, V2 DV2-1) indicate exact transcription, but they are not gate evidence.
- **Portfolio, policy, design, remediation and nonlinear claims:** no EMT evidence either way. They keep their phasor status and their independent ANDES reproduction.
- **Manuscript:** the wording “nonlinear phasor-domain simulation (not EMT)” remains correct and required. Suggested limitations sentence: “*An electromagnetic-transient corroboration in ParaEMT was preregistered four times; the network and operating point were reproduced, but the campaigns stopped at device-interface and modal-measurement qualification gates before any portfolio result, so no EMT evidence is claimed.*”

8 Reproducibility

Preregistration: commit 0555f587. Frozen estimator: commit f8e0d44f.

- `.venv/tx3-analysis EMTSQ1_modal_library.py` — SQ1-1 domain.
- `.venv/xtool-parent EMTSQ1_synth.py check` — generator self-check; runs no estimator.
- `.venv/xtool-parent EMTSQ1_synth.py blind` — SQ1-2.

What EMT did and did not validate (final)

The table separates theorem validity from model-fidelity corroboration.

claim	math	phasor	indep. phasor	EMT (V1+V2+V3+SQ1)	final
V01, V02, V08, V22, V26, T04, T06, T08	proved	yes	where applicable	not applicable	PROVED
V03 P4 trap, $\kappa = 4$ [B02]	—	PCV02; G2 TDS	ANDES Phase 8, 32/32	none (stopped before EMT05)	VALIDATED (model-specific witness)
V06, V07 policy dependence [B01]	structure proved	clean g -only counterfactual	ANDES Phase 8	none	VALIDATED
V10, V11	proved (V10)	PCV04	ANDES R3 11/12	none	PROVED + VALIDATED; VALIDATED (local)
V13, V14	—	PCV04; FC03; G2	partial ANDES	none	VALIDATED (model-specific); BENCHMARK-SPECIFIC as in the ledger
V15–V18; V30, B07, B10	— / normal form	PCV05; G2 TDS	—	none	
V19 network and operating point [I01]	—	reference	ANDES; pandapower after the tap fix	EMT QUANTITATIVELY REPRODUCED (V1 G1)	CONDITIONAL (pandapower translation)
V20 SG dynamics [I02, I03]	—	internal	ANDES R2/R3, E1	V1 G3 FAIL; V2 G3a FAIL; V3 and SQ1 did not reach this gate $\Rightarrow$ UNRESOLVED	VALIDATED (by ANDES)
V21 custom GFL [I04]	—	project DAE	ANDES Phase 8, GATE 5 PASS	V1 G4 FAIL; later campaigns did not reach this gate $\Rightarrow$ UNRESOLVED	VALIDATED as a reproduction of computation; EMT reproduction not achieved
cumulants, closure, V05, V12, B04, B06, negatives	as in the ledger	as in the ledger	—	n/a / none	unchanged